Applied Algebra Qualifying Exam: Part B
Fall 2023

Instructions: Do all problems. All problems are weighted equally. You are not allowed to consult any external resource during this exam. Good luck!

Problem 1: Let G be a finite group and let V be a finite-dimensional G-module over $\mathbb{C}$. Let $\chi : G \to \mathbb{C}$ be the character of V and consider the linear operator $\varphi : V \to V$ given by

$$\varphi(v) := \sum_{g \in G} \overline{\chi(g)}\,(g \cdot v)$$

where $\overline{\chi(g)}$ is the complex conjugate of $\chi(g)$. Assume that V is irreducible.

Prove that there exists a complex number $c \in \mathbb{C}$ such that $\varphi(v) = cv$ for all $v \in V$ and find the value of c.

Problem 2: Find the character table of the alternating group A_4 of even permutations in S_4.

Problem 3: Let G be the product group $G = C_4 \times C_3 \times C_2$, where C_n denotes the cyclic group of over n, and let $\mathbb{F}_3$ be field with 3 elements. Let V be an indecomposable G-module over $\mathbb{F}_3$. Is V necessarily irreducible? Justify your answer.

Problem 4: Let V be the vector space of polynomials $f(x_1, x_2, x_3, x_4, x_5)$ in 5 variables $\{x_1, \ldots, x_5\}$ of homogeneous degree 2 with complex coefficients. The symmetric group S_5 acts on V by variable permutation. Find the decomposition of V into S_5-irreducibles.